

SOC'26

Yash Meshiya

July 2026

1 Report

1.1 Which Algorithm Did What

BFS and **Dijkstra's** are single-pair search algorithms. Each search terminates immediately once the target node has been settled. **bfs** uses a queue data structure to check across all immediate neighbours whereas **Dijkstra's** uses a priority queue which helps in minimizing the cost for reaching target by skipping expensive paths.

A* is implemented in the same way as **Dijkstra's** algorithm but instead of using cost as a parameter we also add a heuristic function which helps in seeing a wider region of search then the immediate neighbour.

BestOrder performs backtracking over all possible coin visitation orders using only the precomputed distance matrix. It does not care about the graph, it just operates on the distance matrix between each pair of keypoints. Any partial solution whose accumulated cost is already greater than or equal to the current best solution is pruned. The function is executed once using the **BFS** distance matrix (uniform-cost mazes) and once using the **Dijkstra's** distance matrix (weighted mazes).

1.2 A* vs. Dijkstra

A* expands only nodes satisfying

$$g(n) + h(n) \leq \text{optimal cost},$$

whereas **Dijkstra** is equivalent to using $h(n) = 0$. Since the heuristic is non-negative, every node expanded by A* is also one that **Dijkstra** would eventually expand.

Maze	Dijkstra	A*	Saved
tiny_2coins	162	96	41%
medium_5coins	1140	571	50%
weighted_4coins	181	140	23%
stress_8coins	4,809	1,779	63%

Table 1: Comparison of cells expanded by Dijkstra and A*

1.3 `weighted_4coins.txt`: Why Weighted Beats Step Count on Cost

This maze contains a low-cost corridor with terrain cost 1 and an expensive column with terrain cost 9.

Examining the path between the two coins reveals that the shortest-step route selected by BFS has a true traversal cost of **36**. In contrast, the minimum-cost path found by Dijkstra avoids the expensive column by returning through the start region, resulting in a traversal cost of only **12**.

Thus, although both routes have similar step counts, the weighted path is three times cheaper in terms of actual traversal cost.

The total optimal solution therefore has a weighted cost of **24**, compared with **22** uniform steps. This does not indicate that the weighted solution is worse; rather, Dijkstra correctly minimizes total traversal cost, even when doing so requires passing through a few cells with costs greater than one.

1.4 Best Coin Order — `medium_5coins.txt`

The optimal coin collection order has a total cost of **32** for both search methods.

Since this maze contains no weighted terrain cells, every move has unit cost. Consequently, Dijkstra reduces to BFS, exactly as in the `tiny_2coins` maze, and both algorithms produce identical optimal solutions.